# Tutorial 7 Mini-Lecture: Exoplanets and Habitable Worlds

## The concepts behind Worksheet 7 • Lectures 13–14

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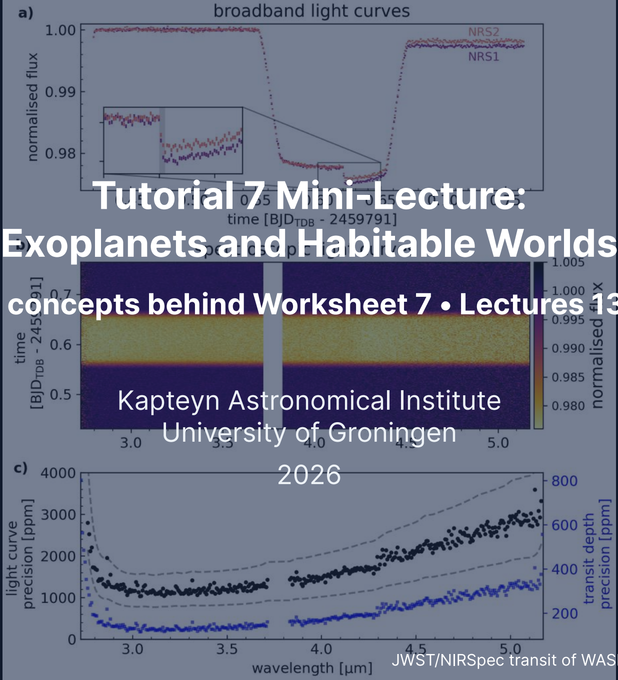
2026




JWST/NIRSpec transit of WASP-39 b. Credit: Alderson et al. (2023)

## What Worksheet 7 trains

Five problems, all built on Lectures 13–14. The physics you need, per problem:

1. **Transits**: the depth $(R_p/R_\star)^2$, the geometric probability $R_\star/a$, why the catalogue is biased.
2. **Radial velocities**: the semi-amplitude $K_\star$, from 51 Peg b to the Earth analogue, mass plus radius gives density.
3. **Transmission spectroscopy**: the scale height, the $2n_H H/R_p$ modulation, flat spectra and how eclipses break them.
4. **The habitable zone**: equilibrium temperature, the Kopparapu flux limits, scaling to K and M dwarfs.
5. **Direct imaging and Drake**: angular separations and contrast, the log-uniform Drake toy model.

Every problem has the shape and level of one exam question. All parts are analytical; no computer is needed.

# Transit depths and geometry

Depth and probability from Lecture 13:

$$\boxed{\delta = \left(\frac{R_p}{R_\star}\right)^2} \qquad p \approx \frac{R_\star}{a}$$

- A hot Jupiter dims a Sun-like star by ~ 1.5%; an Earth analogue by only ~ 84 ppm.
- At 0.05 AU the transit probability is ~ 9%; at 1 AU it drops to ~ 0.5%.

**The catalogue is not the population**

Close-in planets transit more often, and small stars give deeper transits, so the raw catalogue over-represents both. Occurrence rates follow only after the selection bias is modelled out.

Two small numbers, $(R_p/R_\star)^2$ and $R_\star/a$, decide what the transit method can see.

# Radial-velocity masses and densities

The stellar reflex semi-amplitude of Lecture 13:

$$\boxed{K_\star = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_p \sin i}{M_\star^{2/3}}}$$

- 51 Peg b: $K_\star \approx 63\ \mathrm{m\,s^{-1}}$, six times the ELODIE precision.
- An Earth analogue: $\approx 9\ \mathrm{cm\,s^{-1}}$, at the ESPRESSO photon limit but below the activity floor.

**From $m_p \sin i$ to composition**

RV alone gives a minimum mass. A transit forces $\sin i \approx 1$ and adds $R_p$, so mass and radius combine into a bulk density: K2-18 b at ~ $2700\ \mathrm{kg\,m^{-3}}$ cannot be bare rock.

The RV amplitude scales as $m_p P^{-1/3}$; combined with a transit it turns into a density and a first look at composition.

# Scale heights and transmission spectra

The atmospheric lever arm of Lecture 13:

$$\boxed{H = \frac{k_\mathrm{B} T}{\mu\, m_u\, g}} \qquad \frac{\Delta\delta}{\delta} \approx \frac{2 n_H H}{R_p}$$

- Hot Jupiter: $H \approx 630$ km and $\Delta\delta \approx 1120$ ppm, a signal even pre-JWST instruments could see.
- Rocky $\mathrm{CO_2}$ planet: $H \approx 5.8$ km, a factor $\sim 109$ smaller signal.

## When the spectrum is flat

An airless rock, a heavy compact atmosphere, and high clouds all look featureless in transmission. The 15 $\mu$m eclipse of TRAPPIST-1 b (503 K, matching the 508 K bare-rock value) is what broke the tie there: a thick atmosphere is ruled out, a thin one not yet.

Hot and light atmospheres are easy; cold and heavy ones hide, and thermal emission has to break the tie.

# Equilibrium temperature and the habitable zone

Radiation balance and the flux limits of Lecture 14:

$$\boxed{T_{\rm eq} = \left[\frac{L_\star(1 - A_{\rm B})}{16\pi\sigma\epsilon d^2}\right]^{1/4}} \qquad d = \sqrt{\frac{L_\star/L_\odot}{S_{\rm eff}}}\ \mathrm{AU}$$

- Earth: $T_{\rm eq} \approx 255$ K; the 33 K to the observed 288 K is the greenhouse effect.
- The Sun's zone spans 0.97 to 1.69 AU; at TRAPPIST-1 it shrinks to a few hundredths of an AU.

## Small stars, mixed blessing

M dwarfs offer deep transits and long lifetimes, but their habitable-zone planets are tidally locked, baked during the pre-main-sequence phase, and flared for Gyr. K dwarfs may be the best compromise.

The habitable zone moves as $\sqrt{L_\star}$ while stellar lifetime grows as $M^{-2.5}$: the trade-offs are quantitative, not rhetorical.

# Direct imaging and the Drake equation

Small angles and honest priors from Lectures 13–14:

$$\boxed{\theta[\mathrm{arcsec}] = \frac{a[\mathrm{AU}]}{d[\mathrm{pc}]}}$$

- An Earth analogue at 10 pc sits at 0.1 arcsec but at $10^{-10}$ contrast: the contrast is the blocker, not the angle.
- The M-dwarf habitable zone collapses to milliarcseconds: imaging and transits divide the sky between stellar types.

## Drake as a framing tool

With four factors log-uniform over ten decades, $\log_{10} N$ has mean −20 and $\sigma \approx 5.8$: the output restates the priors. “We are alone in the galaxy” is fully consistent with current knowledge.

Direct imaging is a contrast problem; the Drake equation is a map of ignorance, and both statements are quantitative.

## Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** (“show that …”) to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- Most parts close by asking what the number means: that interpretation is graded, not only the arithmetic.

Worksheet and full solutions: `ips.formingworlds.space/worksheets.html`.
We discuss questions, not presentations of the solutions.